FFGFT: Quantum Dots, Synapses and Resonators

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Contents

.1 What a Quantum Dot Is — and Why Shape Matters

Imagine a tiny cavity. Smaller than a dust particle, smaller than a cell, smaller than most viruses — but built from several hundred to a thousand atoms. Trapped inside: one or more electrons, or bound electron-hole pairs.

This cavity is a **quantum dot** (*quantum dot*) — an artificial atom. It behaves like a single atom with sharp, defined energy states — even though it consists of many atoms.

Why? Because it is not the number of atoms that matters. It is the **geometry of the cavity** that matters.

The atoms form only the wall — the **resonator** (a cavity that amplifies certain vibrations and suppresses others). What happens inside is determined by the shape. Just as with a tone chamber in a harmonica: it is not the wood that makes the sound, but the geometry of the channel. The reed vibrates analogously — but the resonator selects discrete frequencies.

Discrete from Outside — Analogue from Inside

A quantum dot has **discrete energy levels** (energy steps that can only take certain specific values, nothing in between) — like notes on a flute. Between two notes there are no further tones on the flute. The jump is sharp.

But the **excitation** (the supply of energy to change the state) is analogue: the intensity of the light, the strength of the electric field, the duration of the pulse — all of this is continuously adjustable.

Discrete and analogue — no contradiction

From outside: continuous excitation, continuous measurement.
From inside: discrete states, sharp jumps.

The analogue and the discrete describe the same physics from two different perspectives. A thermometer shows a continuous temperature — but the atoms inside jump between discrete states.

.2 Geometry Enforces Discreteness

In classical physics a trapped particle can have any arbitrary energy. In a quantum dot this is not possible. There only certain **modes** (permitted vibration patterns in the cavity) are possible — like standing waves in a resonator.

The smallest permitted standing wave fits exactly once into the cavity: half a wavelength from wall to wall. The next fits twice, then three times — and so on. Between these integer multiples there is nothing.

The energies of these modes:

$$E_n = \frac{n^2\pi^2\hbar^2}{2mR^2}, \quad n = 1, 2, 3, \ldots \tag{1}$$

Here R is the radius of the cavity, m the mass of the trapped particle, $\hbar$ Planck's reduced constant, and n the **quantum number**. The factor π^2 follows from the geometry: a standing wave that vanishes at the boundary enforces $k \cdot R = \pi$ for the ground mode, so $k = \pi/R$ and thus $E \sim \pi^2/R^2$. The discreteness comes from the geometry, not from a postulate.

FFGFT Interpretation: Torsion in the Toroidal Field

Within the FFGFT framework an electron is not a point charge in a box — it is a stable **torsion configuration** (a specific twist of the field at a location) of the underlying field.

The discrete levels are the permitted geometric configurations of this field twist in a bounded space. Between the levels there is no stable configuration — the field can only exist in defined patterns, not in arbitrary intermediate states.

FFGFT foundation: Torus geometry enforces discreteness

The universe has in FFGFT the topology of a four-dimensional torus — a closed, self-returning geometry. The boundary condition that generates π^2 is not imposed — it follows from this topology. The factor π is the geometric fundamental constant of the torus. The discreteness is not a quantisation rule that is postulated — it follows from the geometry.

The fundamental parameter of FFGFT is the geometric scaling parameter $\xi = 4/30000 \approx 1.333 \times 10^{-4}$. It generates a universal length hierarchy — from the sub-Planck scale $L_0 = \xi \cdot \ell_P$ up to cosmic distances. Which scales are geometrically distinguished is shown in Section **??**.

Two ξ parameters — two different spaces, no contradiction

In the FFGFT literature two numerically different ξ values appear. This is **not a contradiction** — they describe physically different spaces:

Parameter	Value	Space	Meaning
ξ (geometric)	$\frac{4}{30000} \approx 1.333 \times 10^{-4}$	**3D geometric space**	Length hierarchy of the torus; defines levels L_N; applies in physical position space
ξ_{Higgs} (algebraic)	$\approx 1.038 \times 10^{-5}$	**Number space / quantum field structure**	Corrections in the algebraic field structure; Higgs sector; applies in abstract state space

Why different values? The 3D geometric space and the abstract number space of quantum field theory are different mathematical structures with different boundary conditions. ξ follows from the torus geometry of physical space ($4/30000$ is the ratio of the fundamental geometric length scale to the Planck length in the three-dimensional torus). ξ_{Higgs} follows from the algebraic structure of the Higgs sector in the number space of quantum field theory — a space without direct spatial metric.

In this document: ξ describes the geometric length levels in 3D space; ξ_{Higgs} describes algebraic coupling corrections at the quantum gate level.

.3 Excitons — Bound Pairs as Information Carriers

The most important excited object in the quantum dot is the **exciton** (a bound pair from an excited electron and the hole it leaves behind) — not a free current, but a local displacement in the field.

When light hits the quantum dot, it raises an electron to a higher energy level. In doing so it leaves behind a vacancy — the **hole** (missing electron that behaves like a positive charge). Electron and hole attract each other and form a bound pair: the exciton.

This pair has a characteristic spatial extension radius — the **excitonic Bohr radius**:

$$a_X = \frac{\varepsilon_r \hbar^2}{\mu e^2} \approx 5\text{–}6\,\text{nm} \quad \text{(for CdSe)} \tag{2}$$

If the quantum dot shrinks below a_X, the exciton no longer geometrically fits inside. In FFGFT language: the field torsion needs a minimum cavity to form a stable bound configuration of two counter-rotating torsions.

A quantum dot can carry several excitons simultaneously: $0, 1, 2, 3, \ldots$ — these are discrete storage levels. Each level is sharply distinguishable. A larger quantum dot carries more simultaneous excitons — more distinguishable states in a single object.

.4 Three Classes of Quantum Effects

Before we answer the question of when quantum behaviour sets in and when it disappears, a distinction must be made. There is not *one* quantum effect — there are at least three fundamentally different classes:

Class	Physical basis	Relevant energy	Example
Confinement	Standing waves in the cavity	Level spacing $\Delta E = 3E_1$	Quantum dot, transistor
Tunnelling	Wave function penetrates barrier	Barrier width and height	Enzyme catalysis
Spin coherence	Entangled spin states	Spin relaxation time	Bird navigation

Each class has its own criterion for quantum behaviour. What all three have in common: they need **geometric precision**. Without precise structure — no quantum effect, regardless of the mechanism.

.5 The Two Conditions for Confinement Quantisation

For confinement-based systems — quantum dots, transistors, macroscopic lattices — exactly two conditions apply that *both* must be satisfied:

Two conditions for quantum dot behaviour

Condition 1 — Geometric perfection (absolute, temperature-independent): The resonator must be precise enough to enforce standing waves. A defective lattice shows no confinement quantum behaviour — regardless of how small, regardless of temperature.

Condition 2 — Level spacing vs. heat (size and temperature dependent): The spacing between the lowest two levels must exceed the thermal energy: $\Delta E = 3E_1 > k_B T$.

Condition 1 is the harder one — it is absolute and cannot be replaced by cooling. Condition 2 is softer — cooling shifts the boundary to larger objects.

Only for confinement. Tunnelling and spin coherence follow other criteria (Section **??**).

The Critical Boundary Size

From Condition 2 the maximum resonator size follows directly at which quantum dot behaviour is still possible:

$$\boxed{R_{\text{bulk}}(T) = \pi\hbar\sqrt{\frac{3}{2\,m_{\text{eff}}\,k_B T}}.} \tag{3}$$

Below R_{bulk}: discrete quantum dot behaviour. Above: the levels lie closer than $k_B T$ — band structure, classical behaviour.

Numerical Values

Material	m_{eff}/m_e	R_{bulk} at temperature			
		4 K	77 K	300 K	310 K
GaAs (conduction band)	0.067	221 nm	50 nm	25.5 nm	25.1 nm
CdSe (conduction band)	0.130	159 nm	36 nm	18.3 nm	18.0 nm
Silicon (transverse)	0.190	131 nm	30 nm	15.2 nm	14.9 nm

Cooling to 4 K shifts the boundary for silicon from 15 nm to 131 nm — a factor of ~ 9. This makes macroscopically appearing objects into quantum objects: a 50 nm silicon crystal is classical at room temperature, quantised at 4 K.

The Transition in Detail: Silicon at 300 K

R	E_1	$\Delta E = 3E_1$	$\Delta E/k_B T$	Regime
1 nm	1981 meV	5943 meV	230	strongly quantised
3 nm	220 meV	660 meV	26	strongly quantised
5 nm	79 meV	238 meV	9	transition region
10 nm	20 meV	59 meV	2.3	transition region
15 nm	8.8 meV	26.4 meV	1.0	**boundary**
22 nm	4.1 meV	12.3 meV	0.5	classical
65 nm	0.47 meV	1.41 meV	0.05	classical

The boundary lies at $R \approx 15$ nm — exactly where the classical transistor stopped scaling in 2012 (Section **??**).

Macroscopic Objects: Many Levels, Not None

A macroscopic body shows no quantum dot behaviour — but not because it has no levels, but because it has 10^{23} levels so dense that $k_B T$ covers them all. The discreteness does not disappear. It becomes so fine-grained that it appears as band structure.

Discrete stays discrete — even in the macroscopic

A macroscopic defect-free lattice (Condition 1 satisfied, Condition 2 not) is a quantum object with band structure — not a quantum dot.
A defective lattice (Condition 1 not satisfied) is not a quantum object — regardless of how small, how cold.
The mechanism is always the same: resonator, standing waves, boundary conditions.
Whether 2 states or 10^{23}: the principle does not change.

.6 Biological Quantum Effects: Which Condition Applies Where?

Living cells use quantum effects at 37 °C. Three known examples — and the question of whether the two-condition criterion applies there:

Effect	**Cond. 1: Geometry**	**Cond. 2: $\Delta E > k_B T$**	**Mechanism**
Photosynthesis (FMO)	✓ Protein shell	marginal ($J/k_B T \approx 0.7$)	Exciton coherence; coupling $J \approx 19$ meV
Bird navigation	✓ Cryptochrome	× no confinement ($E_{\text{hf}} \ll k_B T$)	Spin entanglement; relaxation time > reaction time
Enzyme tunnelling	✓ Active site	× no confinement ($E_{\text{zp}}/k_B T \approx 0.05$)	Quantum tunnelling; barrier ~ 0.3–0.5 nm

Photosynthesis: The FMO complex couples chromophores with $J \approx 150\,\text{cm}^{-1} \approx 19$ meV. This is marginal compared to $k_B T(37\,^\circ\text{C}) = 26.7$ meV. Measured coherence times of 300–700 fs at 277 K show that the protein shell acts as a geometrically precise resonator. Current research discusses whether lattice vibrations (phonons) even support the coherence.

Bird navigation: Cryptochrome proteins in the eye produce entangled radical pairs. The relevant energy is the hyperfine splitting $\ll k_B T$ — Condition 2 does not apply here at all. The effect survives because the *spin relaxation time* is longer than the chemical reaction time. This is spin coherence, not confinement.

Enzyme catalysis: Protons tunnel through barriers of ~ 0.3 nm width in the active site. The zero-point energy of the proton lies at $E_{\text{zp}}/k_B T \approx 0.05$ — well below $k_B T$. Tunnelling does not need $\Delta E > k_B T$. It needs only a thin, geometrically precise barrier.

FFGFT finding: Condition 1 is the universal condition

Condition 2 ($\Delta E > k_B T$) applies **only to confinement quantisation**. Tunnelling and spin coherence have their own criteria.
Condition 1 — geometric precision — applies to all three classes without exception.

In FFGFT language: every quantum effect is a stable configuration of the torsion field. Without a precise geometric framework this configuration collapses — whether confinement, tunnelling or spin.

Why 37 Degrees: Geometric Operating Point

The question of the biological operating point at 37 °C has a more precise answer than the usual $k_B T$ argument:

R_{bulk} changes between 0 °C and 57 °C by less than 10 % (from 4.0 nm to 3.6 nm). The collapse at 42–$45\,^\circ$C does not come from $k_B T$ suddenly becoming too large. What collapses is the geometric precision: proteins denature, the torsion configuration breaks down.

Optimal biological operating point

37 °C is not optimal because of a thermal energy limit. It is the point at which biological resonators (protein cavities ~ 2–4 nm, within $R_{\text{bulk}} \approx 3.75$ nm) are maximally chemically active without losing their torsion configuration.
Warm enough for chemistry. Cold enough so that Condition 1 remains stable.

.7 The ξ-Scaling Ladder and the Scale of Quantum Dots

FFGFT has no free length parameter. All physically distinguished scales arise from the iteration of the geometric parameter $\xi = 4/30000$ — beginning at the fundamental length $L_0 = \xi \cdot \ell_P$:

$$L_N = L_0 \cdot \left(\frac{1}{\xi}\right)^N, \quad N = 0, 1, 2, \ldots \tag{4}$$

N	L_N	Physical domain
0	2.15×10^{-39} m	Fundamental FFGFT length (sub-Planck)
1	1.62×10^{-35} m	Planck length ℓ_P
6	3.8×10^{-16} m	Subnuclear regime
7	2.9×10^{-12} m	Atomic scale
8	**2.2×10^{-8} m**	**Nanometres — quantum dot and exciton regime**
9	1.6×10^{-4} m	Cellular scale
10	≈ 1 m	Macroscopic scale

Level $N = 8$ corresponds to $L_8 \approx 22$ nm. The excitonic Bohr radius for CdSe ($a_X \approx 5.5$ nm) lies at the recursion depth

$$N_{\text{exciton}} = \frac{\ln(a_X/L_0)}{\ln(1/\xi)} \approx 7.85. \tag{5}$$

FFGFT finding: $N \approx 8$ as universal scale level

The recursion depth $N \approx 8$ appears in FFGFT at two independent locations:
1. In the geometric derivation of the fine structure constant α: there $N \approx 8$ determines the depth of the fractal self-nesting from which $\alpha \approx 1/137$ follows. *(Derivation in a separate FFGFT document on the fine structure constant; the core statement: the*

fractal self-nesting of the torus to depth $N \approx 8$ yields a length scale $L_8 \approx 22\,\mathrm{nm}$ connected to α via the electromagnetic coupling strength.)
2. At the scale of exciton formation: a_X lies geometrically at $N \approx 7.85$ — the same scale level.
The same geometric hierarchy that determines α also determines at which length scale bound electron-hole pairs can form stable configurations. The scale of excitons is in FFGFT not an empirical quantity — it is geometrically enforced.

The Fractal Correction Factor K_{frak}

The energy level formula applies to an ideally spherical cavity. In FFGFT the spatial geometry is fractal — the effective dimension lies at $D_{\text{frak}} = 2.999867$, not at 3. The universal correction factor:

$$K_{\text{frak}} = \frac{4}{3}\pi \cdot \left(\frac{1}{2}\right)^{3-D_{\text{frak}}} \approx 4.18840, \tag{6}$$

a deviation from the classical sphere volume factor of less than $0.013\,\%$.

FFGFT prediction: fractal shift of QD levels

In FFGFT the energy levels of a quantum dot are systematically shifted compared to the ideal formula. For a CdSe quantum dot with $R = 3\,\mathrm{nm}$:

$$\Delta E_1 \approx 0.0418\,\text{eV} \times 1.3 \times 10^{-4} \approx 5.4\,\mu\text{eV}.$$

This lies near the resolution limit of today's single quantum dot spectroscopy at low temperatures. A systematic measurement of many identical quantum dots could make this offset statistically accessible — as a test of the FFGFT prediction.

.8 Reconfigurable Function — the Same Building Block, Three Roles

Excitation	Function	What happens
Weak continuous light	Memory	Metastable torsion configuration — remains until next excitation
Short strong pulse	Logic gate	Torsion reversal through brief intense excitation
Modulated signal + feedback	Synapse	Weighted forwarding — the weight changes with experience
Electric field	**Stark effect**	Fine tuning of switching thresholds through external field
Magnetic field	**Zeeman effect**	Control via spin states through magnetic field

The same tiny cavity — five different behaviours, depending on what is supplied. No reprogramming. Only physics. The analogue control signal selects a discrete state.

.9 Artificial Synapse — How Hardware Learns

A biological **synapse** (junction between two nerve cells) has three properties that enable learning:

Weight: How strongly does it respond to an incoming signal?

Plasticity: The weight changes with experience. Frequently used connections become stronger — rarely used ones weaker (**Hebbian learning**: neurons that fire together wire together).

Threshold: Below a certain input signal nothing happens at all.

A quantum dot can emulate all of this: The weight corresponds to the **occupation number** (how many of the possible states are currently active) — an analogue value determined by the excitation history. The plasticity arises because repeated excitation permanently shifts the charge state. The threshold is fixed by geometry — but analoguely shiftable through electric fields.

.10 From the Minimal Resonator to the Transistor

The Minimal Resonator: Spin Only

The smallest possible quantum resonator is defined by geometry. Both conditions from Section **??** must be satisfied, and additionally: $E_2 - E_1 \gg k_B T$, so that only the ground mode $n = 1$ is occupied.

For $R = 1.5\,\text{nm}$:

$$E_2 - E_1 \;=\; 3E_1 \;\approx\; 502\,\text{meV} \;\gg\; k_B T(300\,\text{K}) = 26\,\text{meV}. \tag{7}$$

What remains: a single energy level with two permitted orientations — spin-up and spin-down.

The minimum: one torsion configuration, two orientations

A quantum dot with $R \lesssim 1.5\,\text{nm}$ at room temperature realises the **absolute minimum** of a quantum resonator: a single stable torsion configuration of the field (ground mode $n = 1$), with two permitted orientations (spin-up / spin-down). All further quantum systems are extensions of this minimum.

The Hierarchy at the ξ-Level $N \approx 8$

All these systems sit at the same geometric scale level. The transistor makes this particularly clear:

Technology	Channel length	N level	Regime
65 nm node (2005)	65 nm	8.12	classical
22 nm FinFET (2012)	22 nm	**8.00**	quantum influences begin
10 nm node (2016)	10 nm	7.91	tunnelling measurable
5 nm node (2020)	5 nm	7.84	QD regime
3 nm GAA (2023)	3 nm	7.78	genuine QD channel
2 nm (2025)	2 nm	7.73	≈10 atomic layers

When the semiconductor industry *had* to switch to FinFET in 2012 — because classical planar transistors had stopped scaling — the channel lengths had crossed below the ξ level $N = 8.00$. The same level that anchors the excitonic Bohr radius and α geometrically.

Transistor as Extended, Modified Quantum Dot

A modern transistor differs from a simple quantum dot through three deliberate modifications:

Modification	Technical	FFGFT language
Scaling	Channel $L \approx 3$–10 nm, many lattice sites	More permitted modes, transition from spin-only into the multi-level regime
Doping	Deliberate foreign atoms in the lattice	Local torsion perturbations that shift the Fermi level into the band gap — geometrically defined shift, not a random defect
Gate field	External electric field	External torsion field that deforms the resonator geometry of the channel until a new stable configuration locks in

Transistor: quantum dot deliberately modified

A transistor is not a fundamentally different object from a quantum dot. It is an extended resonator at the ξ level $N \approx 8$ with three geometric modifications: **Scaling** gives more states. **Doping** shifts the torsion basis. **Gate field** deforms the resonator geometry from outside.
Modern GAA transistors (3 nm, 2023) have channels of ≈ 10 atomic layers — they are de facto quantum dot arrays. The transistor has returned to its physical origin.

The Complete Hierarchy

System	Scale	States	Characteristic
Spin-only QD	$R \lesssim 1.5$ nm	2	Minimum: one torsion, two orientations
Exciton QD	$R \sim 3$–6 nm	4–20	Bound torsion pairs; memory, logic, synapse
Transistor (classical)	$L \gtrsim 65$ nm	$\gg 1$	Many modes, classical; QD character hidden
Transistor (modern)	$L \sim 2$–5 nm	few	Doped + gate torsion; effectively QD array
Ising machine	optical resonator	2 per unit	Spin-only in the photonic; global feedback

.11 The Arc Closes: the Coherent Ising Machine

The **Coherent Ising Machine** (CIM) — an optical system that solves difficult optimisation problems by physically finding the energetically most favourable state — closes the arc completely.

The basic unit of a CIM is an optical resonator made of mirrors and an **optical parametric amplifier**. This resonator also has discrete states: it oscillates in either one or the other phase — spin-up or spin-down, zero or one. The excitation is analogue — continuous laser light, continuously adjustable coupling strengths. The state is discrete. The same physics, different realisation.

	Quantum dot	Synapse	Ising machine
Resonator	Semiconductor crystal	Protein structure	Optical oscillator
Excitation	Light, E-field	Neurotransmitter	Laser light
States	Energy levels	Weight levels	Spin up/down
Feedback	local	Network	global, all-to-all
Objective	Memory, logic	Learning	Optimisation

The Ising machine is the spin-only quantum dot in the photonic regime — two states per resonator, but with a global coupling network instead of local isolation. From the minimal two-state resonator to the global optimisation machine: the same principle, scaled and connected.

.12 Overall Conclusion: Resonators Everywhere

Conclusion

A quantum dot is a resonator. A synapse is a resonator. A cell is a system of resonators. An Ising machine is a resonator. A transistor is a modified resonator.
All discrete from inside. All analogue from outside. All geometrically precise — otherwise they do not work.
The boundary between classical and quantum mechanical lies neither at a fixed temperature, nor at a fixed size, nor at a material class.
For confinement systems two conditions apply: geometric perfection (absolute) and $\Delta E > k_B T$ (size and temperature dependent). For tunnelling and spin coherence only Condition 1 applies. Condition 1 — geometric precision — is the only universal condition. It applies to all quantum effects.
In FFGFT: all these systems sit at the geometric scale level $N \approx 8$ of the ξ hierarchy — the same level that anchors α and the exciton scale.

Significance for the adaptive hybrid architecture

The hardware toolbox of the adaptive hybrid architecture is a collection of different realisations of the same fundamental principle — deployed according to task: quantum dots for local synaptic operations, Ising machines for global optimisation, macroscopic defect-free lattices for coherent quantum operations.

The learning AI selects — through trial and error — which resonator to use when. It does not learn *on* the hardware. It learns *through* it.

.13 Experimental Confirmation: IBM Quantum Hardware (2025)

IBM Quantum hardware validation of FFGFT foundations, June 2025

The FFGFT foundations developed in this document — in particular the treatment of qubits as torsion configurations with ξ coupling — were validated on genuine quantum hardware:

Hardware: IBM Brisbane & IBM Sherbrooke (127 qubits each), daily calibration, gate fidelity $> 99.9\,\%$ (1-qubit).

Core result: FFGFT amplitudes $E_i(x,t)$ with $\xi_{\text{Higgs}} = 1.038 \times 10^{-5}$ deliver algorithmically equivalent results to standard QM — Bell state fidelity $97.17\,\%$, repetition variance 0.0001–0.001 (excellent).

FFGFT interpretation: The improved repeatability compared to purely probabilistic QM is a direct sign that the deterministic torsion configuration (Section **??**) really exists: the same system under identical conditions delivers more consistent results than the standard collapse picture predicts.

.14 Further Topics in the Follow-up Documents

The following documents build on the foundations of this document:

- **Doc. 174** — *Influencing, Reading and Extending Spin in the FFGFT Framework:* Spin manipulation (Zeeman, EDSR, optical, exchange interaction), spin readout methods (QND measurement as empirical proof of determinism), qudits and extended state spaces.
- **Doc. 175** — *More than Two States — Superposition, Registers, Entanglement in the FFGFT Framework:* Bloch sphere vs. FFGFT cylinder space, Bell states with ξ damping, register entanglement, qudit coding, photonic waveguides.
- **Doc. 176** — *Shor's Algorithm without Qubit Registers — the Photonic Perspective:* QFT as physical wave operation, MZI networks, FFGFT-Shor, eigenvalue problem on sub-Planck lattice, connection to the Riemann zeta function.